

FRUGAL ENGINEERING: AN EMERGING INNOVATION PARADIGM

by [Nirmalya Kumar](#) and [Phanish Puranam](#)

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There may perhaps not be any better example of the dictum that necessity is the mother of invention than can be found in India. Whether it is a refrigerator, ECG device or an automobile, Indian engineers have brought innovative products to market by designing them outside-in. They have looked not only at the needs of the market, but also at what the market can spend. Readers interested in learning how to design more with less will find valuable lessons in this article.

With a vision of mass manufacturing ‘the People’s Car’, Tata Motors set out to design the least expensive production car in the world for under Rupees 100,000 (around \$ 2,000). When it was launched in late 2009, it made headlines around the globe; the Nano was heralded a new breed of transportation. Since then, sales of the Tata Nano have been disappointing, leading many observers to brand it as a failure.

Is the Tata Nano a success or a failure? Did they get the price point wrong? Will it ever really sell outside India? Questions such as these seem — to us — to miss the real story behind innovations like the Nano, GE’s portable ECG machine developed in India, or the ChotuKool (a cheap replacement for a refrigerator targeted to rural Indian consumers). However interesting each of these innovations might be, (or even how much money these particular products may lose or make), it is of much more fundamental interest to recognize the revolution in product-design philosophy that they embody. Carlos Ghosn, who heads Renault-Nissan, is credited with coining the term frugal engineering, to signify achieving more with fewer resources. Others have described similar ideas under terms such as reverse innovation, Gandhian innovation or even “Indo-vention.” Whatever the term, the real story behind the Nano is a radically new product-design philosophy and a new approach to innovation. (See article on Reverse Innovation elsewhere in this issue).

Global leadership in an industry is often linked to having demanding domestic market consumers. It is so partially because Japanese consumers are extremely demanding when it comes to electronics, German consumers for automobiles, Americans for fast food products, and French and Italian consumers for fashion, so much so, in fact, that one observes leading companies in these industries from the vantage point of their respective countries and the needs of their respective markets.

A similar dynamic is alive and well in India. Innovation is a direct response to meeting the needs of Indian marketplace, where consumers are both demanding as well as budget constrained. According to Guillermo Wille, former Managing Director at GE India, “the beauty of the Indian market is that it pushes you in a corner...it demands everything in the world, but cheaper and smaller.”

In our research, we found that India was building a capability for frugal engineering. Given that this appears to be a practice that is taking root in China and other emerging markets, we conducted extensive research on frugal engineering. We discovered many examples that in turn led us to identify six underlying principles or pillars on which such frugal engineering efforts often seem to rest. We describe these principles in this article.

1. Robustness

India is a harsh environment in terms of the huge variances that occur in operating conditions in that country. Such a setting affects the priorities that drive product development and innovation. This means not just extremes of temperature, but an erratic supply of electricity and peak-load ratios unheard of in the West.

Nokia has gained a dominant share of the Indian market and has hired more people there than in any other country except for its home base of Finland. Each of Nokia’s three Indian R&D centres is an integral part of the firm’s global R&D infrastructure. Product features developed for the Indian market include a dust-proof keypad and face, a torchlight, and nonslip sides for a better grip during humidity, the bane of the monsoon season. It is quite conceivable that these are features that a Finnish inventor in stable, cool Nordic conditions might never imagine.

2. Portability

Space constraints and the need to transport products to rural areas over poor transportation links highlight the importance of portability in India. Small and lightweight become highly desirable product attributes. Think again of the MAC 400i ECG, and consider the miniaturization efforts that are necessary to achieve this kind of portability in a product class that typically ascribes no value at all to portability.

3. Defeating

Defeating consists of feature rationalization, or “ditching the junk DNA” that tends to accumulate in products over time. With Indian consumers, firms can avoid implementing features that do little to enhance the actual product. Siemens supports its innovation efforts in India with a billion-euro investment devoted entirely to developing and adapting products for the local market. Currently, Siemens employs seventeen thousand people who work on 42 products that eventually will be sold in India and exported to other markets. Most of the prototypes are developed and held in strict secrecy, but leaks hint at the possibility of solar-powered X-ray machines, fetal heart monitors, steam turbines, and road traffic management systems. The CEO

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of Siemens, Peter Löscher, alluded to the potential: “These products require a different kind of innovation. What counts here is simplicity, not sophistication.”¹

4. Leapfrog technology

It may seem a contradiction, but some infrastructure gaps in India have positively affected Indian innovation: they have forced entrepreneurs and companies to adopt technologies that make relying on existing infrastructure (creaking and unreliable as it is in many ways) simply irrelevant. Indian engineers have invented a battery-powered, ultra-low-cost refrigerator resistant to power cuts; an automatic teller machine for rural areas; and even a flour mill powered by a scooter. People in the West, with its constant access to electricity, have little motivation to pursue such innovations. The Indian mobile phone industry is the poster child for leapfrogging over infrastructural constraints. A limited fixed-line infrastructure created an opportunity for mobile phones to reach many more people. Mobile telephony is also relatively cheap, sharable, and easily repaired. And thus, a new frontier of global innovation opened in India.

5. Megascale production

With its massive population, India has market segments that if captured, can help firms drive costs down so that they can produce on a massive scale—megaproduction. The costs of component manufacturing for the Nano, for alternative energy products, or for mobile handsets in India, similarly reflect the scale at which these products can be produced. The health sector in India is another example, where having one billion plus potential patients leads to process revolutions that lower the costs dramatically compared to prevailing world prices. The low-cost cataract operations performed at the Aravind Eye Hospital are a prime example of this.

6. Service Ecosystems

The conventional wisdom in marketing is that reaching demanding cost targets requires low variability, or a one-size-fits-all mentality. Selling large volumes requires that a product appeal to multiple segments, each with slightly different needs. Displaying innovative thinking, Indian firms resolve the dilemma by using efficient service ecosystems. While these ecosystems achieve low costs, they also highlight product features and thus broaden the product's appeal. Today, it is easy to see a plethora of small repair shops and other businesses that have mushroomed around population centres in India. The use of these service ecosystems—which comprise not just parts and repair but financing as well—can help firms enlarge their product markets. For example, Selco, which designs simple, low-cost systems that combine solar panels and storage batteries, has installed solar lighting systems in 100,000 homes in rural southern India. A two-light home system can cost around \$200, or 10,000 rupees—expensive, considering that the company's customers earn less than half that in a month. So Selco assembled an aggressive financing package with various local rural banks that provide financing to 85 percent of Selco customers; the repayment rate for solar loans is about 90 percent. In addition, service support personnel visit customers once every three months during the first year to ensure that the system is working properly and to collect batteries for recycling.

For companies in emerging markets, developing innovative products and launching them across the world, especially for end consumers, powerfully enhances corporate reputation and brand equity. However, as NASA's space program demonstrates, when the focus is on exploring the frontiers of science without a budget constraint, the United States is unmatched in its innovation capabilities. Therefore, it is not surprising that the majority of the Nobel prizes awarded for science go to academics at U.S. universities doing basic research. Such basic research develops new knowledge, which may or may not have any immediate real world applications. It is often curiosity driven and funded by university or government grants in order to build the long-term competitive advantage of the country. It is hoped that over time, some percentage of this new knowledge created will result in breakthrough products and services. India, and even China, despite the latter's intense focus to do so, cannot easily compete with developed countries in this domain in the foreseeable future.

BELLS AND WHISTLES VS. THE UNADORNED, STRIPPED DOWN

Let us now turn to innovation that is made with the intention of generating a profit, as is the case with new product development by companies. As an example, consider the iPad, which caused as great a frenzy as any new product one can remember in recent times.² One million iPads sold in the first 28 days after the launch, requiring the international launch to be delayed because Apple was unable to keep up with the demand in the United States. Interestingly, the iPad was not a new product developed based on careful market research. Rather it was dreamt up by Apple to satisfy unarticulated needs of consumers. Buyers in the United States had to pay \$499 for the basic version of the iPad. Most probably, these eager buyers already owned a \$1,000-plus laptop computer and a \$100-plus mobile smart phone. In addition, the ongoing cost of connecting all of these devices to the Internet would have easily amounted to \$100 or more per month. The point is that, again, the West is excellent at developing iPad-type innovative products that ignite the desire to purchase in rich (relative to a global scale). Indian and Chinese companies struggle to find an advantage in launching such end products for consumers who are so far away in physical, economic, emotional and experiential terms from their domestic markets.

In contrast to this iPad universe of rich consumers in the developed world, consider the larger but much different world of budget constrained consumers of India and other emerging markets. Here the mobile phone is the device of choice for billions of consumers. With 5 billion mobile subscriptions worldwide, more people have access to the cell phone than to a clean toilet. In India, telecom companies like Bharti Airtel and Reliance routinely sell handsets for less than \$25. These cheap handsets are then connected to the network with no ongoing monthly charges. One-cent-a-minute phone calls across India, one-cent text messages, and cheap access to the web open up a world of possibilities. Notwithstanding that they have the cheapest mobile phone tariffs in the world, Indian telecom companies have earned huge profits.

Contrasting the iPad with the cheap mobile phone services demonstrates that the developed world and the emerging markets have moved on different trajectories. The developed world's innovators are building for an ever-expanding bandwidth network and spiralling toward fancier, costlier, more network-hungry and status-giving devices. In contrast, emerging market innovators are constantly seeking new uses for the cheap and basic mobile phone, which are used for banking, weather forecasts, market reports, and finding employment. And the developed world's domestic demand for ever-sleeker, faster, fancier devices makes it harder for them to innovate for the larger, much-less affluent world outside, one still dominated by frugal wants. It is in this domain where Indian innovation can make a difference.

Best practices in frugal engineering are unlikely to be found in a textbook. It is up to sophisticated Indian engineers and managers in companies to create a set of standards, a body of knowledge, and a community of practitioners around the concept, as it has been created for quality management or the Toyota production system. Policy makers in India can help encourage research and training in codified frugal-engineering practices, once these have developed sufficiently. A well-developed body of frugal engineering practices that can be applied to meet the needs of budget-constrained consumers is the ultimate gift of the Nano, not merely the offer of a car with a less than 100,000-rupee price tag.

¹ James Lamont, "The Age of Indovation Dawns," Financial Times, 15 June, 2010

² Anand Giridharidas, "Innovation without bells and whistles," International Herald Tribune, 10 April 2010, p. 2

The Authors:

Nirmalya Kumar

Nirmalya Kumar is professor of marketing and Phanish Puranam is professor of strategy and entrepreneurship at London Business School. They are also co-directors of the Aditya Birla India Centre at the school. This article is adapted from their book India Inside: The emerging innovation challenge to the West (Harvard Business Review Press 2012).

Phanish Puranam

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